Calculus

Bobby

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Contents

Real Numbers and Limits of Sequences

Problem 1

Suppose that $\{a_n\}_{n=1}^{\infty}$ is a sequence of real numbers converging to a real number L. Determine the value of the following limit:

$$\lim_{n\to\infty}\left(\frac{a_1+2a_2+\cdots+na_n}{1+2+\cdots+n}\right)$$

You are required to prove your conclusion using ϵ-N language.

Hint 1

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Hint 2

Decompose the numerator into two parts:

$$\sum_{k=1}^{n} kb_k = \underbrace{\sum_{k=1}^{N} kb_k}_{\text{Fixed finite sum}} + \underbrace{\sum_{k=N+1}^{n} kb_k}_{\text{Controlled by } \epsilon}$$

Problem 2

Let $c \in (0, \frac{3}{4})$ and $x_1 \in (0, 1)$. Define the sequence $\{x_n\}_{n=1}^{\infty}$ by the recurrence relation $x_{n+1} = 1 - cx_n^2$. Given that all terms of $\{x_n\}_{n=1}^{\infty}$ are distinct, prove that the sequence $\{x_n\}$ converges and find its limit.

Hint 1

Let $f(x) = 1 - cx^2$. Note that $f(x)$ is decreasing on $(0, 1)$. Therefore, the sequence $\{x_n\}$ is not monotonic. Consider the monotonicity of the composite function $g(x) = f(f(x))$.

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Hint 2

Investigate the odd terms $\{x_{2n-1}\}$ and even terms $\{x_{2n}\}$ separately. Since $g(x)$ is increasing, these subsequences are monotonic. Show that they are bounded and thus convergent.

Hint 3

Let $x_{2n} \to \alpha$ and $x_{2n-1} \to \beta$. We have:

$$\begin{cases} \beta = 1 - c\alpha^2 \\ \alpha = 1 - c\beta^2 \end{cases}$$

Subtract the two equations: $\alpha - \beta = c(\alpha^2 - \beta^2)$.

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Hint 4

Factor the difference to get $(\alpha - \beta)[1 - c(\alpha + \beta)] = 0$. Assume $\alpha \neq \beta$. Derive a contradiction using the given condition $c < 3/4$.

Problem 3

Given that $\lim_{n\to\infty}(2x_{n+1}+x_n) = A \in \mathbb{R}$. Prove that the sequence $\{x_n\}$ converges.

Hint 1

Let $y_n = 2x_{n+1} + x_n$. We are given that $y_n \to A$. Rewrite the recurrence relation to express x_{n+1} in terms of x_n:

$$x_{n+1} = -\frac{1}{2}x_n + \frac{1}{2}y_n$$

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Hint 2

Assume $\lim_{n\to\infty} x_n = L$ exists. Taking the limit on both sides of the equation from Hint 1:

$$L = -\frac{1}{2}L + \frac{1}{2}A$$

Solve for L.

Hint 3

Let $L = A/3$. Consider the distance $|x_{n+1} - L|$. Use the Triangle Inequality to show:

$$|x_{n+1} - L| \leq \frac{1}{2}|x_n - L| + \frac{1}{2}|y_n - A|$$

Since $\frac{1}{2} < 1$ and $|y_n - A| \to 0$, deduce that $|x_n - L| \to 0$.

Problem 4

Calculate the limit of the sequence:

$$\lim_{n\to+\infty}\left(\frac{\ln 1+2\ln 2+3\ln 3+\cdots+n\ln n}{n^2}-\frac{\ln n}{2}\right).$$

Hint 1

Let $S_n = \sum_{k=1}^{n} k \ln k$. First, combine the two terms into one fraction to apply the **Stolz Theorem**:

$$\lim_{n \to +\infty} \frac{S_n - \frac{1}{2} n^2 \ln n}{n^2}$$

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Hint 2

When calculating the difference of the numerator, you will encounter the term $(n-1)^2 \ln(n-1)$. Rewrite it using logarithms:

$$\ln(n-1) = \ln\left[n\left(1 - \frac{1}{n}\right)\right] = \ln n + \ln\left(1 - \frac{1}{n}\right)$$

Hint 3

To evaluate the limit, use the second-order Taylor expansion for $\ln(1-x)$ as $x \to 0$:

$$\ln\left(1-\frac{1}{n}\right) = -\frac{1}{n} - \frac{1}{2n^2} + o\left(\frac{1}{n^2}\right)$$

Substitute this back to simplify the numerator difference.

Continuity of Functions

Problem 1

Let $a_1, a_2, \ldots, a_n$ be arbitrary positive numbers. Calculate the limit:

$$l = \lim_{x \to 0} \left(\frac{a_1^x + a_2^x + \cdots + a_n^x}{n} \right)^{\frac{1}{x}}.$$

Hint 1

Recall the standard result: if $\lim_{x\to 0} \alpha(x) = 0$, then

$$\lim_{x\to 0} [1+\alpha(x)]^{\frac{1}{x}} = e^{\lim\limits_{x\to 0} \frac{\alpha(x)}{x}}$$

In this problem, let the base be $1+\alpha(x)$, so:

$$\alpha(x) = \frac{a_1^x + \cdots + a_n^x}{n} - 1 = \frac{\sum_{i=1}^{n}(a_i^x - 1)}{n}$$

Hint 2

To evaluate the exponent limit $\lambda = \lim_{x\to 0} \frac{\alpha(x)}{x}$, rewrite it as:

$$\lim_{x\to 0} \frac{1}{n} \sum_{i=1}^{n} \frac{a_i^x - 1}{x}$$

Use the standard limit (definition of derivative at $x = 0$):

$$\lim_{x\to 0} \frac{a^x - 1}{x} = \ln a$$

The final answer involves the sum of logarithms.

Derivatives and Differentials of Functions

Problem 1

Calculate the following limit:

$$\lim_{x\to 0} \frac{1-\cos x\sqrt{\cos 2x}\,\sqrt[3]{\cos 3x}}{x^2}.$$

Hint 1

Recall the second-order approximations:

$$\cos u \approx 1 - \frac{1}{2}u^2$$

$$(1+u)^\alpha \approx 1 + \alpha u$$

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Hint 2

Apply the approximations to the composite functions.

$$\sqrt{\cos 2x} = (\cos 2x)^{\frac{1}{2}} \approx \left(1 - \frac{1}{2}(2x)^2\right)^{\frac{1}{2}} \approx 1 - x^2$$

Do the same for $\sqrt[3]{\cos 3x}$, then multiply the linear approximations.